WATER RESOURCES ENGINEERING

Subject Code: 23A01501 | UNIT – V : DIVERSION HEAD WORKS

1. Introduction – Head Works

Head works = structures constructed at the head (start) of a canal to regulate the supply from a river/ reservoir into the canal system. They raise the river level, divert water into the canal, control silt & floods.

❖Types of head works

- (A) Storage head works: dam/ reservoir built across a river to store monsoon flow and release as needed (Nagarjuna Sagar, Srisailam). Used for non-perennial rivers, hydro-power, flood control.
- (B) Diversion head works: weir or barrage across a perennial river which raises (pounds) the water level so that it flows by gravity into the canal (Dowleswaram barrage on the Godavari, Prakasam barrage on Krishna at Vijayawada).

❖Diversion vs storage head works

Point	Diversion	Storage
Purpose	Raise level, divert flow	Store water for later use
Structure	Weir/ barrage (low height)	Dam (high)
River	Perennial, assured flow	Seasonal flow
Storage	Very little (pond)	Large reservoir
Cost	Low	Very high
Silt	Problem (silting of pond)	Settled in reservoir
Examples	Prakasam, Dowleswaram, Sukkur	Srisailam, Hirakud

2. Weirs and Barrages

Weir = a solid obstruction (raised crest) across a river with fixed crest (no gates, or only a few scouring gates), raising water level by the height of the crest; water flows over the crest in floods and the level is maintained by crest height (upstream pond above crest, with shutters on the top).

Barrage = a diversion structure with a low-crest and large gated openings (bays) with piers across the river; pond level is controlled by gates (raised/ lowered) – no (or small) permanent crest; flood passes through the gates (water level in flood regulated by gate opening).

❖Types of weirs (by section)

- Vertical drop (masonry) weir – vertical downstream face; water falls into a water cushion (on the rear) – for hard rock/ strong foundations.
- Rock-fill weir with sloping aprons – loose rock, sloping upstream & downstream aprons; cheap in boulder rivers (Punjab, rocky beds).
- Concrete weir with (sloping) glacis – modern – concrete floor with glacis (slope 3:1 to 2:1), hydraulic jump forms on glacis/ at the toe dissipating energy: Type A (jump on the floor), Type B (jump on the glacis), Type C (jump at downstream of glacis) – adopted on permeable (alluvial) foundations – most common.
- Cut-off wall / sheet-piles, impervious floor provide seepage safety.
- Drum/ rubber (inflatable) weirs; shutter weirs.

❖Weir vs barrage

Feature	Weir	Barrage
Crest	High, fixed crest	Low crest, gates control level
Pond control	Crest height + shutters	Gates
Flood passage	Over crest (afflux ↑)	Through gate openings (less afflux)
Cost	Lower (less steel/ gates)	Higher (gates, piers, bridge)
Silt control	Less	Better (gates flush silt)
Afflux & submergence	Higher	Lower
Road bridge	Not usually	Often combined (traffic bridge over piers)
Examples	Anicut (Tamil Nadu), Okhla (old)	Sukkur, Farakka, Prakasam, Dowleswaram, Dhavaleswaram

3. Layout of Diversion Head Works & Components

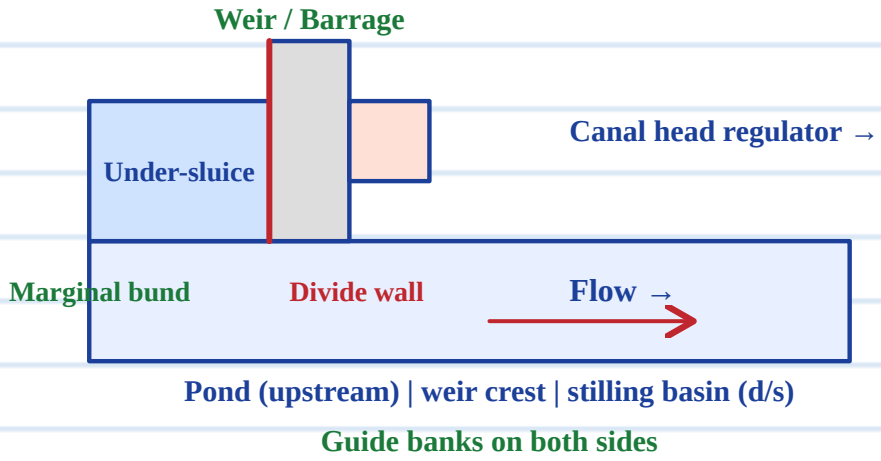


Fig 5.1 Layout of diversion head works (plan, simplified)

❖General layout (from upstream to downstream, plan)

- River → upstream marginal bunds & guide banks → weir/ barrage (main river section) + under-sluices (scouring sluices) on the canal side → divide wall → canal head regulator at the off-take, silt excluder in front of regulator, fish ladder, river training works, downstream (stilling basin/ energy dissipation), cut-offs & aprons.

❖Components

- 1. Weir/ barrage proper: raises the pond level; crest & gates; impervious floor (concrete), glacis, piers.
- 2. Under-sluices (scouring sluices): gated openings at the canal end of the weir (on the canal side), with crest level lower than the weir crest (usually at river bed level); purposes: (i) scour the silt deposited in front of the canal head regulator, (ii) maintain a clear, deep channel towards the head regulator, (iii) pass low floods without opening weir gates, (iv) regulate the river.
- 3. Divide wall: masonry/ concrete wall (RC) separating the under-sluice portion from the weir and providing a still pocket for silt excluders; perpendicular to the weir axis; extends up to the beginning of the upstream pond & beyond the downstream floor; prevents cross-flow from the weir into the under-sluice pocket; cross-currents are controlled.
- 4. Fish ladder: a passage (step-pool) for fish migration across the weir – a channel with baffles/ steps (1 in 10 slope), near divide wall.

- 5. Canal head regulator: gated structure at the head of the canal controlling the discharge entering the canal (also prevents entry of silt & flood water); crest level ≥ 1.2 m higher than under-sluice crest; aligned at 90° – 110° to the weir axis (to reduce silt entry).
- 6. Silt control devices: Silt excluders (in front of canal head – separate silt-laden bottom layer & divert it into the under-sluices; tunnels under head regulator bay) and silt ejectors (extractors) (on the canal – just beyond the head regulator – remove the silt that has already entered the canal; tunnel type).
- 7. River training works: guide banks (guide the river flow axial to the weir, trumpet shaped, length ≥ 1.0–1.5 × width of waterway, prevent river from outflanking), marginal (afflux) bunds (protect the upstream land/ banks from submergence due to afflux; each side, freeboard 1.5 m above HFL + afflux), spurs/ groynes (to deflect flow, protect banks).
- 8. Pond/ Upstream pocket: the pool (pondage) upstream of the weir; pond level = crest level + shutters or gate level; afflux = rise in HFL due to the weir.
- 9. Floor, aprons, cut-offs (sheet piles), protection works: upstream & downstream impervious floor, stilling basin, toe walls/ cut-off, inverted filter, launching apron (loose stone/ cc blocks) on both ends, upstream blocks.
- 10. Trash rack, stop-log gates, service bridge/ gate-operating deck.

❖Location of head works

- Straight, stable reach of river, with good rock/ clay foundation, high banks & minimum cross-drainage; narrow river width; above the command area (commanding level); near the canal alignment; good approach for construction.

4. Failure of Hydraulic Structures on Permeable Foundations – Causes

> On pervious (sandy/ alluvial) foundations, the weir is subject to seepage (water percolating under the structure) from the upstream pond to downstream; this gives uplift pressure and exit gradients, causing failures.

❖A. Failure due to subsurface flow

- (i) Piping (undermining): seepage water emerges at the downstream end with a velocity (hydraulic gradient > critical), carrying away fine soil grains ("boiling/ quicksand" condition), forming a pipe-like channel ("pipe") backward under the floor → progressive undermining, floor settles, collapse.
  - Critical (exit) gradient $i_c = (G_s - 1)(1 - n) \approx 1.0$ ($G_s = 2.65$, $n = 0.4 \rightarrow 0.99$); safe gradient is 1/5 – 1/7.
- (ii) Rupture of floor by uplift pressure: uplift (upward water pressure) on the underside of the floor is more than the weight of floor + water above → floor cracks/ bursts (uplift is highest at upstream end of floor) → seepage increases → failure.

❖B. Failure due to surface flow

- (iii) Scour/ erosion: high-velocity flow from the structure scours the river bed & banks downstream (retrogression) & upstream of the pier ends, undermining the floor ends.
- (iv) Suction caused by standing wave (hydraulic jump): negative pressure leads to removal of floor blocks.
- (v) Rupture by hydraulic jump (impact) if the jump forms on the floor without enough thickness; excess velocity damages the floor/ stilling basin.
- (vi) Failure by overturning/ sliding of the weir body (not due to seepage – instability).
- (vii) Impact of floating debris, silting, downstream retrogression of bed level.

❖Remedies (safety measures)

- Provide upstream & downstream cut-offs (sheet piles/ concrete) – increase the creep path & reduce exit gradient, cut-off at downstream (deeper) to protect against piping.
- Provide a longer impervious floor/ blanket; proper floor thickness (against uplift); inverted filter + pervious floor (weep holes) with launching aprons on the downstream end (to relieve uplift & stop piping).
- Stilling basin with baffle blocks; proper energy dissipation; protect toe with CC blocks & loose stone aprons; pressure-relief drainage holes/ pipes.
- Provide "pervious floor" downstream & "impervious floor" upstream.

5. Seepage Theories – Design of Impervious Floor

- Seepage under the weir follows curved path (streamlines) from upstream cut-off/ floor to the downstream exit; the head H (upstream pond level – downstream tail water level) is lost along the path.

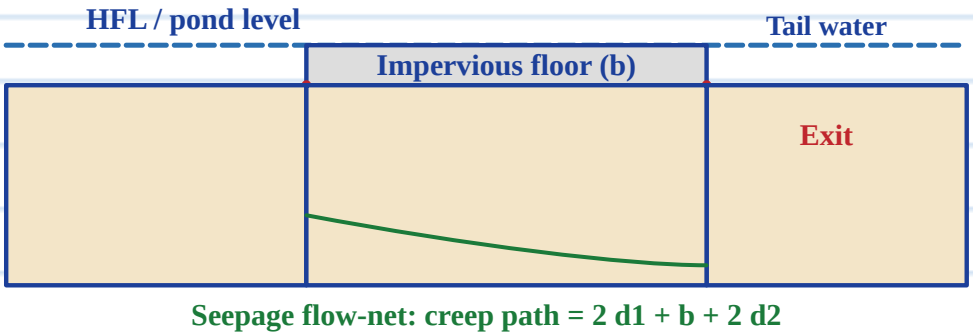


Fig 5.2 Impervious floor with upstream & downstream cut-off piles (seepage path)

- Design requirements: (i) safe exit gradient (no piping); (ii) floor thickness safe against uplift. Floor length & cut-off depths are fixed by seepage theories (Bligh – Lane – Khosla).
- Theories: Bligh's creep theory (1910), Lane's weighted creep, Khosla's method (1936 – flow net/ mathematical solutions – scientifically most accurate).

6. Bligh's Creep Theory

> Bligh: the water seeping under a weir creeps along the contact of the floor and the foundation (along the surface of the structure); the head loss is proportional to the length of creep (percolating water loses head at a constant rate along its path). Total creep length $L = L_h + L_v$ (horizontal + vertical, with no distinction).

- Creep length $L = b + 2\,d_1 + 2\,d_2 + \ldots$ = total (horizontal floor length b + twice the depth of each pile/ cut-off (down & up)). (Slopes steeper than 45° counted as vertical; flatter counted as horizontal.)

❖Assumptions

- Loss of head ∝ creep length (uniform hydraulic gradient H/L); vertical and horizontal creep are equally effective; the water particle follows the structure surface; no effect of pile positions/ concentrations; no change of gradient with depth.

❖Safety against piping

$$L = C\,H \rightarrow \text{hydraulic gradient } H/L = 1/C \text{ (must be} \leq 1/C\text{)}$$

- C = Bligh's coefficient (depends on soil; finer soil → larger C → longer creep path needed).

Type of soil	Bligh's C	Safe gradient 1/C
Light sand & mud	18	1/18
Fine micaceous sand	15	1/15
Coarse grained sand	12	1/12
Gravel & sand (boulders)	9	1/9
Boulders, shingle, gravel & sand	4 – 9	1/4 – 1/9

- Rule: provided creep length $\geq C\,H$.

❖Safety against uplift (floor thickness)

- Residual (uplift) head at a point at creep distance l from the upstream end:

$$h_l = H - (H/L)\cdot l \text{ (m of water)}$$

- Uplift pressure $p = \gamma_w h_l$ (acting on the underside); the floor must resist this by its own weight:

$$t = (4/3)\cdot h/(G-1),\ G = \text{specific gravity of floor material } (\approx 2.24 \text{ for concrete})$$

- Derivation: $(t + h_{top})\,\gamma_w$ ... weight of floor per unit area in water = $(G-1)\,\gamma_w\,t \geq \gamma_w\,h \rightarrow t = h/(G-1)$; extra 1/3 (safety) → $t = (4/3)\,h/(G-1)$. For $G = 2.24$: $t \approx 1.075\,h$.
- Floor must be thick enough at the points where h is greatest (upstream end); thickness increases from the downstream end towards where uplift is greater.

❖Limitations / defects of Bligh's theory

- Does not distinguish between horizontal & vertical creep (vertical piles are more effective!); no distinction between outer & inner faces of piles.
- Neglects position of cut-offs (a downstream pile is more effective than the same pile upstream); uniform hydraulic gradient (loss of head ∝ length) is wrong – actual loss is non-uniform (rapid near ends).
- Does not check safety against exit gradient (the true cause of piping) – only a "safe" average gradient; no uplift pressure variations/ mutual interference of piles considered.
- Provides approximate safe results for small works only; overdesigned or unsafe for pile at the end.
- Lane's weighted creep theory (improvement): $L_w = \Sigma L_v + (1/3)\Sigma L_h \geq C_w H$; C_w = 8.5 (very fine sand/ silt), 7.0 (fine sand), 6.0 (medium sand), 5.0 (coarse sand), 4.0 (fine gravel), 3.0 (coarse gravel/ soft clay), 2.5 (boulders), 1.8 (hard clay).

❖Example – Bligh

> A weir has floor 40 m long with upstream cut-off 3 m and downstream cut-off 4 m deep; head H = 4.5 m; soil: coarse sand C = 12.

- Creep length L = 40 + 2×3 + 2×4 = 54 m; required = CH = 12 × 4.5 = 54 m ✔ (just safe).
- Uplift at point A (d/s end of floor, before the d/s pile): creep up to A = 3 + 3 + 40 = 46 m → h = 4.5 – (4.5/54)(46) = 0.667 m; t = (4/3)(0.667/1.24) = 0.72 m.
- Uplift at toe of u/s pile: l = 3 → h = 4.5 – 0.25 = 4.25 m.

7. Khosla's Theory

> Khosla (1936): Seepage under a structure occurs in curved stream-lines and equipotential lines forming a flow net; the flow is governed by Laplace's equation ($\nabla^2\varphi = 0$). Khosla & co-workers solved the equation (Joukowski/ conformal transformation) for simple standard profiles and gave the pressure at key points and the exit gradient. Complex profiles are broken into simple standard forms. Results are expressed as percentage of head (φ).

- Key points: the junctions of the floor with pile lines and the bottom of piles (points E, D, C) in simple profiles; upstream floor start = 100%, downstream tail = 0%.
- Concept: loss of head isn't proportional to the creep length; piles (cut-offs) are much more effective at the ends; downstream pile controls the exit gradient, upstream pile reduces the uplift on the floor.

❖Standard profiles (Khosla's simple forms)

- (a) Straight floor with a pile at the upstream end (U/S pile).
- (b) Straight floor with a pile at the downstream end (D/S pile).
- (c) Floor with an intermediate pile.
- (d) Floor depressed/ raised (sloping), floor with two piles → solved by combining & correcting these.

❖Notation

- b = total floor length (horizontal); d = depth of the pile; $\alpha = b/d$; $\lambda = [\,1 + \sqrt{(1+\alpha^2)}\,]/2$; H = maximum seepage head; φ = pressure expressed as % of H (residual).

❖Pressure percentages (pile at the downstream end, i.e., b/d profile)

$$\varphi_E = (1/\pi)\cos^{-1}[(\lambda-2)/\lambda]\quad \varphi_D = (1/\pi)\cos^{-1}[(\lambda-1)/\lambda]\quad \varphi_C = 0$$

- (angle in radians; result × 100 gives %; i.e., $\varphi(\%) = \cos^{-1}(\ldots)$(degrees) ÷ 1.8)

❖Pile at the upstream end

$$\varphi_{E1} = 100\%\quad \varphi_{D1} = 100 - \varphi_D\quad \varphi_{C1} = 100 - \varphi_E$$

- (use the same φ_E, φ_D expressions with $\alpha = b/d$ for the u/s pile).
- Intermediate pile: pressures at the points E, D, C of the pile are read from Khosla's curves (depending on $\alpha = b/d$ and the position of pile b_1/b), or by the same type of expressions with modified α.

❖Exit gradient

> Exit gradient G_E = hydraulic gradient of the seepage at the downstream end of the floor (exit), the upward gradient at the toe; if it exceeds the critical value (≈ 1) → piping.

$G_E = (H/d) \cdot [1/(\pi\sqrt{\lambda})]$ (d = depth of downstream cut-off)

- With λ as above ($\alpha = b/d$). For a long floor (large α): $G_E \approx 1.41\, H/(\pi\sqrt{b\,d})$.
- Safe exit gradients (Khosla):

Soil	Safe G_E
Shingle (gravel)	1/4 – 1/5
Coarse sand	1/5 – 1/6
Fine sand	1/6 – 1/7
Critical gradient (all)	≈ 1 ($i_c = (G_s - 1)(1 - n)$)

- Condition: $G_E \leq$ safe value → increase d (deeper d/s cut-off), or b, to reduce G_E. Factor of safety 4–6.
- Exit gradient charts: G_E d/H plotted vs $1/\alpha$.

8. Corrections to Khosla's Pressures

- Since the actual profile is a combination of piles and floor, the pressures read for a simple profile need correction:
- (1) Correction for mutual interference of piles: piles influence each other; correction $C\,(\%) = 19\sqrt{(D/b')} \times (d + D)/b$
  - D = depth of the influencing pile line (the pile whose effect on another pile is being found); d = depth of the pile on which the influence is calculated; b' = distance between the two piles; b = total floor length.
  - Sign: positive (+) for points in the rear (behind, i.e. upstream side) of the influencing pile; negative (–) for points ahead of it in the direction of flow.
- (2) Correction for floor thickness: key points E, C are at the top surface of the floor in Khosla's curves; actual pressure is needed at the bottom of the floor; interpolate linearly between the pile-bottom and floor-top values: (E' = pressure at the bottom of floor) corrected: E: negative, C: positive correction ($\Delta\Phi = (\varphi_D - \varphi_E)\, t/d$).
- (3) Correction for slope of floor: $C_s = C \times b_s / b_1$ (Khosla's table of C for slope 1:1…1:6 – positive for downward slope (in the direction of flow), negative for upward slope; applies to the pressure at the key point at the start/ end of the slope).

❖Steps in design by Khosla's method

- 1. Fix H, soil type & safe exit gradient; choose d/s cut-off depth d (below scour level, ≥ 1.2–1.5 × scour depth) and floor length b (initially from Bligh/ Lane).
- 2. Compute $\alpha = b/d$, λ; $G_E = H/(d\pi\sqrt{\lambda})$; compare with safe value; modify d or b if needed.
- 3. Find φ_E, φ_D, φ_C for each pile (standard form); apply corrections (interference, thickness, slope).
- 4. Residual uplift head at each key point = (φ%/100) × H – (head above floor/ hydrostatic); thickness of floor $t = h/(G - 1)$ (or $4/3 \times h/(G-1)$); fix t at key points; provide a nominal minimum (0.5–0.6 m) under the crest**.
- 5. Check overturning/ sliding; design aprons, filters.

❖Example – Khosla (D/S pile)

> Floor b = 30 m, d/s pile d = 3 m, H = 5 m. $\alpha = 30/3 = 10$; $\lambda = [1 + \sqrt{(1 + 100)}]/2 = (1 + 10.05)/2 = 5.525$.

- $\varphi_E = (1/\pi)\cos^{-1}[(5.525 - 2)/5.525] = \cos^{-1}(0.638)/180° = 50.4/180 = 28.0$ % → pressure head = 0.28 × 5 = 1.40 m.
- $\varphi_D = \cos^{-1}[(5.525 - 1)/5.525 = 0.819]/180 = 35.0/180 = 19.4$ % → 0.97 m.
- Exit gradient $G_E = (5/3) \times 1/(\pi \times \sqrt{5.525}) = 1.667 \times 0.1354 = 0.226 \approx 1/4.4$ → > safe 1/6 (fine sand) ✗ → increase d to ≈ 5 m: $\alpha = 6$, $\lambda = (1+\sqrt{37})/2 = 3.54$; $G_E = (5/5) \times 1/(\pi \times 1.88) = 0.169 \approx 1/5.9$ ✔ (safe for coarse sand).

9. Determination of Uplift Pressure & Impervious Floor Length

❖By Bligh

- Total creep $L = C\,H$ gives minimum floor length: $b = C\,H - 2\Sigma\, d$ (with cut-offs).
- Uplift head at any point: $h = H - (H/L)\,l$; floor thickness $t = (4/3)\,h/(G - 1)$ (with G = 2.24).
- Weak point of Bligh: only the average; may be unsafe against piping because exit gradient is not checked.

❖By Khosla

- Uplift at key points obtained from φ values (after corrections); residual head at floor point = $\varphi \times H$; net uplift = $h_{residual}$ – (hydrostatic head above floor point); thickness $t = h_{net}/(G - 1)$; exit gradient G_E check.
- More realistic & recommended; flow net verified by electrical/ sand analogy and numerical (FEM).

❖Comparison

Point	Bligh	Khosla
Basis	Empirical, creep length	Flow net/ potential theory (Laplace)
Head loss	Uniform along creep length	Non-uniform, depends on profile
Vertical vs horizontal creep	Equal	Different effectiveness
Pile position/ depth	Ignored	Accounted (u/s pile reduces uplift; d/s pile reduces exit gradient)
Exit gradient	Not checked	Calculated & checked
Uplift pressure	Linear variation	Accurate (curved) distribution
Corrections	None	Interference, thickness, slope
Accuracy	Approximate	High – adopted for design
Use	Preliminary/ small works	Final design

10. Solved Problems

❖Problem 1 – Bligh

Fine sand $C = 15$, $H = 3$ m: creep length required $L = 45$ m; with piles 2 m + 3 m → floor $b = 45 - 2(2+3) = 35$ m.

❖Problem 2 – Exit gradient

$H = 4$ m, $d = 2.5$ m, $b = 25$ m: $\alpha = 10$, $\lambda = 5.525$ → $G_E = (4/2.5)/(\pi \times 2.35) = 0.217$ ($> 1/6$ ✗). With $d = 4$ m, $\alpha = 6.25$, $\lambda = (1+\sqrt{40.06})/2 = 3.66$ → $G_E = 1/(\pi \times 1.913) = 0.166$ ($\approx 1/6$) ✔.

❖Problem 3 – Uplift & thickness (Khosla)

φ_E = 28% at E, H = 5 m → residual pressure head = 1.4 m. Taking the net uplift = 1.4 m (conservative, ignoring water cover on the floor): t = 1.4/(2.24 – 1) = 1.13 m; with the 4/3 safety factor t ≈ 1.5 m.

❖Problem 4 – Floor thickness by Bligh

Residual head at a point = 2.5 m, G = 2.24 → t = (4/3)(2.5/1.24) = 2.69 m.

❖Problem 5 – Critical gradient

G_s = 2.65, n = 0.40 → i_c = 1.65 × 0.6 = 0.99 ≈ 1; safe with FOS 5 = 0.2 = 1/5.

II. Important Questions – Unit V

- Types of diversion head works; differentiate diversion and storage; weir vs barrage.
- With a neat layout, explain the components of diversion head works (under-sluices, divide wall, fish ladder, head regulator, silt excluder/ ejector, guide banks).
- Causes of failure of hydraulic structures on permeable foundations & remedies.
- Bligh's creep theory: assumptions, design for piping & uplift; limitations.
- Khosla's theory: standard profiles, pressures φ_E, φ_D, φ_C, corrections; exit gradient; safe values.
- Compare Bligh's & Khosla's theories.

Key formulas: $L = C\,H$; $t = (4/3)\,h/(G-1)$; $\lambda = [1+\sqrt{(1+\alpha^2)}]/2$; $G_E = H/(d\pi\sqrt{\lambda})$; Safe G_E: 1/5 – 1/7